14 y is inversely proportional to d^2
When $d = 10$, $y = 4$

d is directly proportional to x^2
When $x = 2$, $d = 24$

Find a formula for y in terms of x.
Give your answer in its simplest form.

Direct proportion: $d = kx^2$

1 mark
Write the direct relationship

The x2 column means x^2.

x2	d
4	24
1	6

(÷4 from 4 to 1; ÷4 from 24 to 6)

$24 = k \times 2^2 = 4k$

$k = 6$, so $d = 6x^2$

1 mark
Find a constant from the given values

Inverse-square proportion: $y = \frac{K}{d^2}$

1 mark
Use inverse square

$4 = \frac{K}{100}$ so $K = 4 \times 100 = 400$.

Substitute $d = 6x^2$:

$y = \frac{400}{(6x^2)^2} = \frac{400}{36x^4}$

1 mark
Combine both relationships

Divide numerator and denominator by 4:

$y = \frac{100}{9x^4}$

1 mark
Simplified formula $y = \frac{100}{9x^4}$

(Total for Question 14 is 5 marks)

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